

§ 100. Ideal classes according to the orders.

The concept of equivalence of ideals and the theorems on class numbers can be presented most clearly if, according to Vol. II, § 169, one represents the ideals by means of the functionals discussed in Vol. II, § 153 ff., since in this way one may use the ordinary rules of multiplication and division.¹

We restrict ourselves here once and for all, in the body Ω , to numbers and ideals which are relatively prime to an arbitrarily assumed whole rational number Q , the conductor of an order.

Under this supposition the whole and fractional numbers of the body Ω form the group previously denoted by O . We enlarge this group by adjoining the functionals, and denote the group so enlarged by $\overline{O}$. To each element of $\overline{O}$ there corresponds then a whole or fractional ideal, and to such elements whose quotient is a functional unit there corresponds the same ideal.

If η is a number in O and ε is any functional unit, then the product $\varepsilon\eta$ corresponds to a principal ideal. Accordingly we denote

$\overline{E}$ the group of functional units,

E the group of numerical units,

and by

$$\overline{EO}$$

the principal class of the functionals.

If we take a system of representatives

$$\varphi_1, \varphi_2, \dots, \varphi_h \tag{1}$$

of non-equivalent whole functionals in Ω , then every whole or fractional functional Φ can be represented in the form

$$\Phi = \varphi_i \omega,$$

where ω is a functional from $\overline{EO}$; and the ideal classes in Ω are the cosets of $\overline{EO}$ in $\overline{O}$. Hence the class number h is the index of the divisor $\overline{EO}$ of $\overline{O}$:

$$h = (\overline{O}, \overline{EO}). \tag{2}$$

Here, in the sharper division into classes, one must in O restrict oneself to the numbers of positive norm.

If now we consider the division of the ideals into classes according to the order $[Q]$, we must regard the functional group $\overline{EO}'$ as principal class, and the class number is

$$h' = (\overline{O}, \overline{EO}'). \tag{3}$$

In order to derive from this the relation between h and h' , we make use of the general group theorems of Vol. II, § 2, which we formulate and supplement as follows.

12. *Theorem.* If A is a group and B a divisor of A of finite index (A, B) , and furthermore C a divisor of B of finite index (B, C) , then, by Vol. II, § 2, (4),

$$(A, C) = (A, B)(B, C). \tag{4}$$

Let A again be a group with the divisor B of index μ . We put

$$A = a_1 B + a_2 B + \dots + a_\mu B, \tag{5}$$

¹Compare also the author's memoir "Über Zahlengruppen in algebraischen Körpern", first communication, *Mathematische Annalen* 48, 438.

where $a_1, a_2, \dots, a_\mu$ is a complete system of representatives of A modulo B .

Let now C be a group of elements which can be composed with the elements of A (for example, in our case simply by multiplication). Then from (5) there follows

$$AC = a_1BC + a_2BC + \dots + a_\mu BC. \quad (6)$$

Now one and the same element can occur in the two cosets a_1BC , a_2BC only if $a_1a_2^{-1}$ is contained in BC , but not in B .

From this we formulate the theorem:

13. *Theorem.* If A is a group, B a divisor of A of finite index (A, B) , and C a group composable with A , of such a kind that A and BC contain no common elements except those of B , then

$$(AC, BC) = (A, B). \quad (7)$$

The supposition of this theorem may also be expressed as follows:

B is the intersection of A and BC ;

and it is fulfilled, for example, when C is contained in B , and hence $B = BC$, and also when C has no element in common with A .

14. *Theorem.* If A is a group composed from the two groups A' and B ,

$$A = A'B, \quad (8)$$

and if B' is the intersection of A' and B , then

$$(A, B) = (A', B'), \quad (9)$$

provided these indices are finite.

For by (8) there is contained, in every coset aB of B in A , an element a' from A' , and one can therefore also represent these cosets by $a'B$, where a' is contained in A' . If then a'_1B, a'_2B are different cosets modulo B in A , then a'_1B', a'_2B' are different cosets modulo B' in A' , and conversely; for $a'_1a'^{-1}_2$ is contained in B' if and only if it is at the same time contained in B , and from this formula (9) follows.

We now apply these theorems to the expressions (2) and (3) for h and h' . By (4),

$$(\overline{O}, \overline{EO}') = (\overline{O}, \overline{EO})(\overline{EO}, \overline{EO}'), \quad (10)$$

and therefore

$$h' = h(\overline{EO}, \overline{EO}'). \quad (11)$$

We now denote, as above, by E the group of numerical units in O . Then $\overline{E}E = \overline{E}$, since the numerical units are contained among the functional units; and every number which is contained in $\overline{EO}'$ is contained in EO' . Hence EO' is the intersection of O and $\overline{EO}'$. If, therefore, we apply (7), putting $O, EO', \overline{E}$ in place of A, B, C , then B is the intersection of A and BC , and we obtain

$$(\overline{EO}, \overline{EO}') = (\overline{EO}, \overline{EEO}') = (O, EO'). \quad (12)$$

Furthermore, by (4),

$$(O, EO')(EO', O') = (O, O'), \quad (13)$$

and if by E' we denote the group of the numerical units in O' , that is, the intersection of E with O' , so that $E'O' = O'$, then

$$(EO', O') = (EO', E'O') = (E, E')$$

by (9), since E' is the intersection of $E'O'$ and E . Therefore from (13) we have

$$(O, EO')(E, E') = (O, O'),$$

and from (12)

$$(\overline{EO}, \overline{EO}')(E, E') = (O, O'),$$

and finally, from (11),

$$(E, E')h' = (O, O')h. \quad (14)$$

15. *Theorem.* Formula (14) remains unaltered if O' denotes not precisely the order group, but any number group contained in O , of finite index (O, O') , and if we call two ideals $\mathfrak{a}, \mathfrak{a}'$ equivalent according to O' when their quotient $\mathfrak{a}'/\mathfrak{a} = \eta'$ is a number in O' . If E' is the group of the units contained in O' , and if (E, E') has a finite value, then the class number h' is finite and is determined by formula (14).

For example, if O is the group of all numbers in Ω (except 0), and O' the group of numbers of positive norm, then, if there are any numbers of negative norm at all, $(O, O') = 2$. If there are then units of negative norm, then also $(E, E') = 2$. But if all units have positive norm, then $(E, E') = 1$, and we obtain in the first case $h' = h$, in the second case $2h' = h$.

The equivalence according to O is then the general equivalence; that according to O' is the sharpened equivalence.

We return to the orders $[Q]$, and under O understand the numbers relatively prime to Q (if necessary only those of positive norm). Then (O, O') has been determined in the preceding paragraph.

If the discriminant D and its root Δ are negative, then in general both E and E' contain only the two units ± 1 . In the two exceptional cases $\Delta = -3, -4$, E contains six and four units, respectively; but E' , when $Q > 1$, contains only two; and therefore $(E, E') = 1$ in general, $(E, E') = 3$ in the case $\Delta = -3$, $(E, E') = 2$ in the case $\Delta = -4$. Thus in these cases, by § 98, (10), we have

$$\lambda h' = Q \prod_q \left(1 - \frac{(\Delta, q)}{q}\right) h, \quad (15)$$

where

$$\lambda = 2 \quad \text{for } \Delta = -4, \quad \lambda = 3 \quad \text{for } \Delta = -3, \quad \lambda = 1 \quad \text{for } \Delta < -4. \quad (16)$$

This is the relation which was first derived by Gauss and later confirmed by Dirichlet in another way.

If, for example, $\Delta \equiv 1 \pmod{4}$ and $Q = 2$, then, according to Gauss's terminology, h is the class number for the forms of the second kind, h' that of the forms of the first kind. It is

$$\begin{aligned} Q \left(1 - \frac{(\Delta, 2)}{2}\right) &= 1, \quad \Delta \equiv 1 \pmod{8}, \\ &= 3, \quad \Delta \equiv 5 \pmod{8}. \end{aligned}$$

Thus

$$\begin{aligned} h' &= h, \quad \Delta \equiv 1 \pmod{8}, \\ h' &= 3h, \quad \Delta \equiv 5 \pmod{8}, \end{aligned}$$

and only in the exceptional case $\Delta = -3$ does $h' = h$ occur.

If D is positive, it remains to determine (E, E') . Let

$$\varepsilon = \frac{t + u\sqrt{D}}{2}$$

be the fundamental unit in O , and let λ be the smallest positive exponent for which

$$\varepsilon^\lambda = \frac{t + u\sqrt{D}}{2}$$

is contained in O' . Then E consists of the powers $\pm\varepsilon^\nu$, and E' of the powers $\pm\varepsilon^{\lambda\nu}$, and it is

$$(E, E') = \lambda,$$

and formula (15) gives the class number h' . A further determination of λ cannot in general be given.

If again $\Delta \equiv 1 \pmod{4}$ and $Q = 2$, then $\lambda = 1$ or $= 3$; namely $= 1$ when in (17) u is even, and $= 3$ when u is odd. The latter can occur only when $\Delta \equiv 5 \pmod{8}$; consequently, for $\Delta \equiv 1 \pmod{8}$ one always has $\lambda = 1$, while for $\Delta \equiv 5 \pmod{8}$ λ can be both 1 and 3.